

Unit-2

CSS

Style Sheets

Need Of CSS

- CSS is a language that describes the style of an HTML document.
- CSS describes how HTML elements should be displayed.
- **CSS** to define how HTML elements should appear.

CSS Selectors

1. Element Selector
2. Id Selector
3. Class Selector
4. Group selector

Element Selector

- The element selector select the element based on element name

```
p
{
Background-color:red;
}
```

Id selector

- Id selector use the id attribute of the HTML element to select specific element.

```
#abc
```

```
{
```

```
background-color:red;
```

```
}
```

Class selector

- The class selector element based on class attribute.

.abc

{

Background-color:red;

}

Group selector

- If we have elements with same style.

p

```
{ background-color:red;}
```

h1

```
{ background-color:red;}
```

we can write above code as

p,h1

```
{ background-color:red;}
```

What is CSS?

- **CSS** stands for **Cascading Style Sheets**
- CSS describes **how HTML elements are to be displayed on screen.**
- **CSS saves a lot of work.** It can control the layout of multiple web pages all at once.**[With an external stylesheet file, you can change the look of an entire website by changing just one file!]**
- External style sheets are stored in **CSS files.**

CSS Backgrounds

- CSS background properties:

1. background-color
2. background-image
3. background-repeat
4. background-attachment
5. background-position

Background Color

- The background-color property specifies the background color of an element.

Example

```
body
{
    background-color: lightblue;
}
```

Background Color

Example 2

- Example

```
h1 {  
    background-color: green;  
}  
  
div {  
    background-color: lightblue;  
}  
  
p {  
    background-color: yellow;  
}
```

Background Image

- The background-image property specifies an image to use as the background of an element.
- By default, the image is repeated so it covers the entire element.

```
body {  
    background-image: url("paper.gif");  
}
```

Repeat Horizontally or Vertically

```
body
{
    background-image: url("gradient_bg.png");
    background-repeat: repeat-x;
}
```

Background-repeat values:

repeat-x

repeat-y

no-repeat

Background-position

```
body {  
    background-image: url("img_tree.png");  
    background-repeat: no-repeat;  
    background-position: right top;  
}
```

- It has three different types of values:

Length values (e.g. 100px 5px)

Percentages (e.g. 100% 5%)

Keywords (e.g. Right top)

Background-position

- The default values are 0 0. This places your background image at the top left of the container.
- The first value is the horizontal position, second value is the vertical position.
- So **100px 5px** will move the image 100px to the right and five pixels down.

Background-position

- Keywords are just shortcuts for percentages. It's easier to remember and write top right than 0 100%, and that's why keywords are a thing. Here is a list of all five keywords and their equivalent values:

top: 0% vertically

right: 100% horizontally

bottom: 100% vertically

left: 0% horizontally

center: 50% horizontally if horizontal isn't already defined. If it is then this is applied vertically.

Background Image - Fixed position

```
body
{
    background-image: url("img_tree.png");
    background-repeat: no-repeat;
    background-position: right top;
    background-attachment: fixed;
}
```

Shorthand Property **Background**

body

{

background: #ffffff url("img_tree.png") no-repeat fixed right top;

}

Sequence:

background-color

background-image

background-repeat

background-attachment

background-position

CSS Borders

Different border Properties

- border-width:medium/thick/thin
- border-style (required)
- border-color
- Border-collapse: initial/seperate/collapse

Border Width

- The border-width property specifies the width of the four borders.

one

```
{  
    border-style: solid;  
    border-width: 5px;  
}
```

Border Style

The border-style property specifies what kind of border to display.

The following values are **allowed**:

- dotted - Defines a dotted border
- dashed - Defines a dashed border
- solid - Defines a solid border
- double - Defines a double border
- groove - Defines a 3D grooved border. The effect depends on the border-color value

Border Style

- ridge - Defines a 3D ridged border. The effect depends on the border-color value
- inset - Defines a 3D inset border. The effect depends on the border-color value
- outset - Defines a 3D outset border. The effect depends on the border-color value
- none - Defines no border
- hidden - Defines a hidden border

Border Color

The border-color property can have from one to four values (for the top border, right border, bottom border, and the left border).

```
.one {  
    border-style: solid;  
    border-color: red;  
}
```

Border - Individual Sides

- From the examples above you have seen that it is possible to specify a different border for each side.

```
p {  
    border-top-style: dotted;  
    border-right-style: solid;  
    border-bottom-style: dotted;  
    border-left-style: solid;  
}
```

Border - Shorthand Property

```
p {  
    border: 5px solid red;  
}
```

Sequence

- border-width
- border-style
- border-color

CSS Margins

- The CSS margin properties are used to generate space around elements.
- The margin properties set the size of the white space outside the border.
- CSS has properties for specifying the margin for each side of an element:
 1. `margin-top`
 2. `margin-right`
 3. `margin-bottom`
 4. `margin-left`

Example

```
p {  
    margin-top: 100px;  
    margin-bottom: 100px;  
    margin-right: 150px;  
    margin-left: 80px;  
}
```

Margin - Shorthand Property

```
p {  
    margin: 100px 150px 100px 80px;  
}
```

Sequence

- margin-top
- margin-right
- margin-bottom
- margin-left

Margin - Shorthand Property

margin: 100px 150px 100px 80px;

Top Right Bottom Left

margin: 100px 150px 100px;

Top Right/Left Bottom

margin: 100px 150px;

Top/bottom Right/left

Margin: 100px;

All

CSS Padding

- The CSS padding properties are used to generate space around content.
- The padding clears an area around the content (inside the border) of an element
 1. padding-top
 2. padding-right
 3. padding-bottom
 4. padding-left

Example

- ```
p {
 padding-top: 50px;
 padding-right: 30px;
 padding-bottom: 50px;
 padding-left: 80px;
}
```

# Padding - Shorthand Property

```
p {
 padding: 50px 30px 50px 80px;
}
```

## Sequence

- padding-top
- padding-right
- padding-bottom
- padding-left

# Padding - Shorthand Property

padding: 100px 150px 100px 80px;

Top Right Bottom Left

padding: 100px 150px 100px;

Top Right/Left Bottom

padding: 100px 150px;

Top/bottom Right/left

padding: 100px;

All

# CSS Text

## TEXT CSS Properties

- color
- Text-alignment: center/inherit/justify/left/right
- text-decoration:blink/inherit/line-through/none/overline/underline
- Text-transform:capitalize/inherit/lowercase/uppercase
- Text Indent
- Letter Spacing
- Line-Height
- Direction:rtl/ltr
- Word-spacing
- Text-shadow: *h-shadow v-shadow color* | none | initial | inherit;

# Text Color and Text Align

```
h1 {
 color: green;
}
```

```
h1 {
 text-align: center/left/right;
}
```

# Text Decoration

```
a {
 text-decoration: none;
}
```

## Other Values for Text Decoration

overline

line-through

underline

# Text Transformation

The text-transform property is used to specify uppercase and lowercase letters in a text.

```
p {
 text-transform: uppercase;
}
```

- Other values should be  
uppercase  
lowercase  
capitalize

# Text Indentation

- The text-indent property is used to specify the indentation of the first line of a text:

```
p {
 text-indent: 50px;
}
```

Output

In my younger and more vulnerable years my father gave me some advice that I've been turning over in my mind ever since. 'Whenever you feel.

# Letter Spacing & Line Height

```
h1 {
 letter-spacing: 3px;
}
```

```
p {
 line-height: 0.8;
}
```

# Text Direction and Word Spacing

- The direction property is used to change the text direction of an element:

```
div {
 direction: rtl;
}
```

The word-spacing property is used to specify the space between the words in a text.

```
h1 {
 word-spacing: 10px;
}
```

# Text Shadow

- The text-shadow property adds shadow to text.

The following example specifies the position of the horizontal shadow (3px), the position of the vertical shadow (2px) and the color of the shadow (red):

```
h1 {
 text-shadow: 3px 2px red;
}
```

CSS FONT

# CSS Font Family

- The font family of a text is set with the font-family property
- Example

```
p {
 font-family: Times New Roman, Times, serif;
}
```

# Font Style

- The font-style property is mostly used to specify italic text.
- This property has three values:
- normal - The text is shown normally
- italic - The text is shown in italics
- oblique - The text is "leaning" (oblique is very similar to italic, but less supported)

# Font Style

- Example

```
p {
 font-style: normal;
}
```

# Font Size

```
h1 {
 font-size: 40px;
}
```

Set Font Size With Em [emphemeral unit]

*pixels/16=em*

*e.g. 16pixels is 1em.*

# Font Weight

- The font-weight property specifies the weight of a font:

```
p {
 font-weight: normal/bold;
}
```

# Font Variant

- The font-variant property specifies whether or not a text should be displayed in a small-caps font.
- In a small-caps font, all lowercase letters are converted to uppercase letters. However, the converted uppercase letters appears in a smaller font size than the original uppercase letters in the text.

# Font Variant: Example

```
p{
 font-variant: normal;
}
```

```
h1 {
 font-variant: small-caps;
}
```